IIR Filters

⇒ **IIR** :

$$h(n) = a^n u(n)$$

Such $h(n)$ has infinite or very large values.

⇒ IIR filters are digital filters so instead of designing it directly, we use analog filters.

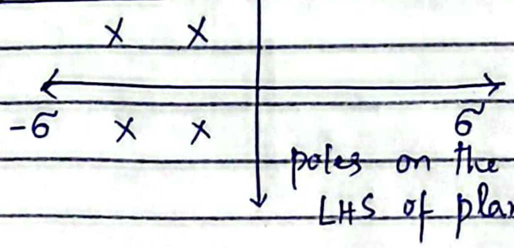
goal

$\boxed{\text{Analog}} \Rightarrow$ Mapping $\Rightarrow$ digital filter

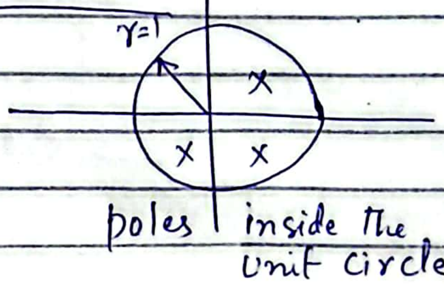
So, we will map the characteristics of analog filter onto digital filters.

Analog filters	Digital filters.
Laplace domain	Z-domain
$s = j\Omega$	$z = e^{j\omega}$
$(\sigma = 0)$	$(r = 1)$
Ω = analog freq.	ω - digital freq.
rad/sec	rad
$-\infty \leq \Omega \leq \infty$	$-\pi \leq \omega \leq \pi$

To stable



To stable



⇒ To design digital filter from analog filter, stability factor should be considered. Systems stable in one domain should be stable in other domain.

⇒ It means poles on the LHS of $j\Omega$ should be inside the unit circle of $j\omega$ axis.

⇒ So, I will consider few conditions while mapping from S-plane to Z-plane.

1. poles in the LHS of S-plane must be mapped ~~on~~ to poles inside the unit circle of Z-plane.

2. poles on $j\Omega$ axis → onto unit circle.

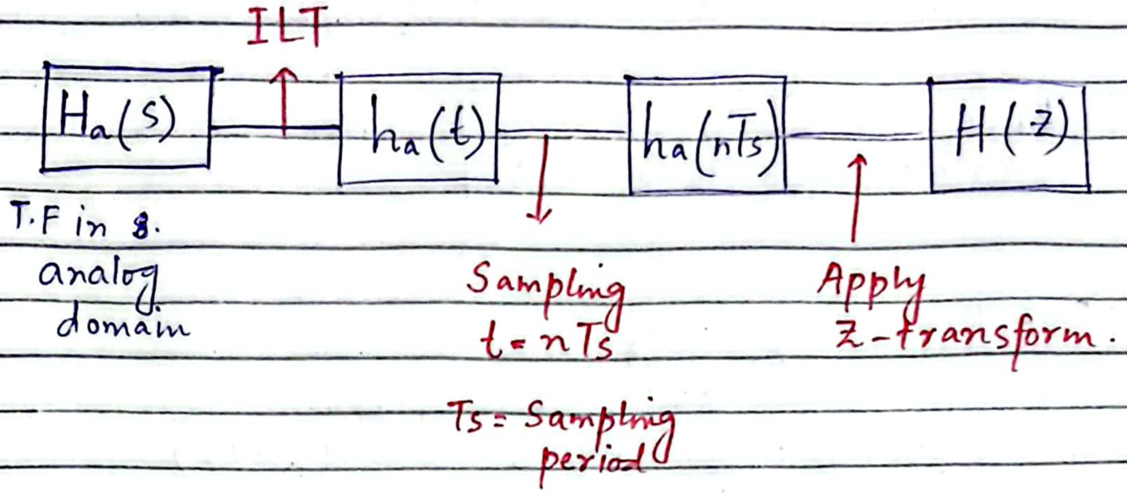
3. poles on RHS of S-plane → outside the unit circle.

-: Mapping Techniques :-

(i) Impulse invariance Method (IIM)

(ii) Bilinear transformation Method (BLT)

-: Impulse Invariance Method:-



Let us Consider

$$H_a(s) = \sum_{k=1}^{N} \frac{A_k}{s-p_k}$$

Step I: Apply ILT

$$L^{-1}\{H_a(s)\} = L^{-1}\left\{\sum_{k=1}^{N} \frac{A_k}{s-p_k}\right\}$$

$$= L^{-1}\left\{\frac{A_1}{s-p_1} + \frac{A_2}{s-p_2} + \dots \frac{A_N}{s-p_N}\right\}$$

$$= L^{-1}\left\{\frac{A_1}{s-p_1}\right\} + L^{-1}\left\{\frac{A_2}{s-p_2}\right\} + \dots L^{-1}\left\{\frac{A_N}{s-p_N}\right\}$$

$$= A_1 L^{-1}\left\{\frac{1}{s-p_1}\right\} + A_2 L^{-1}\left\{\frac{A_2}{s-p_2}\right\} + \dots A_N L^{-1}\left\{\frac{A_N}{s-p_N}\right\}$$

As we know

$$L^{-1}\left\{\frac{1}{s-a}\right\} = e^{at}u(t)$$

$$= A_1 e^{p_1 t} u(t) + A_2 e^{p_2 t} u(t) + \dots A_N e^{p_N t} u(t)$$

$$\boxed{h(t) = \sum_{k=1}^{N} A_k e^{p_k t} u(t)}$$

Step II :- $t = nT_s$

$$h(nT_s) = \sum_{k=1}^{N} A_k e^{p_k (nT_s)} u(nT_s)$$

Step III :- Apply Z-transform

$$H(z) = \sum_{n=-\infty}^{\infty} h(nT_s) z^{-n}$$

$$H(z) = \sum_{n=0}^{\infty} \left\{ \sum_{k=1}^{N} A_k e^{p_k (nT_s)} \right\} z^{-n}$$

$$H(z) = \sum_{k=1}^{N} A_k \left[\sum_{n=0}^{\infty} e^{p_k (nT_s)} z^{-n} \right]$$

$$= \sum_{k=1}^{N} A_k \left[\sum_{n=0}^{\infty} \left(e^{p_k T_s} z^{-1} \right)^n \right]$$

Recall, $\sum_{n=0}^{\infty} a^n = \frac{1}{1-a}$.

$$= \sum_{k=1}^{N} A_k \left[\frac{1}{1 - e^{p_k T_s} z^{-1}} \right]$$

$$H(z) = \sum_{k=1}^{N} \frac{A_k}{1 - e^{P_k T_s} z^{-1}} \quad \text{— (1)}$$

We start from

$$H(s) = \sum_{k=1}^{N} \frac{A_k}{s - P_k} \quad \text{— (2)}$$

from eq. 2 poles are $s = p_k$.

" eq. 1 poles are $z = e^{P_k T_s}$

$\therefore \boxed{z = e^{sT_s}}$ relation b/w eq (1) & (2)

Criteria for mapping :-

$$z = e^{sT}$$

$s = \sigma + j\Omega$

$z = re^{j\omega}$

$re^{j\omega} = e^{(\sigma + j\Omega)T}$

$re^{j\omega} = e^{\sigma T} \cdot e^{j\Omega T}$

Compare real & Img parts

$\boxed{r = e^{\sigma T}}$ $\boxed{\omega = \Omega T}$

If $\sigma < 0$

$r < 1$

If $\sigma = 0$

$r = 1$

If $\sigma > 0$

$r > 1$

Exp1:-

$$H_a(S) = \frac{1}{(S+1)(S+2)}$$

$f_s = 5$ samples/sec.

Solution:-

$H_a(s) \longrightarrow h(t) \longrightarrow h(nT_s) \longrightarrow H(z)$

Step I :- Apply ILT to obtain $h(t)$

$$\frac{1}{(S+1)(S+2)} = \frac{A}{S+1} + \frac{B}{S+2}$$

After p.f

$$H_a(s) = \frac{1}{S+1} + \frac{(-1)}{S+2}$$

$$\therefore \ h_a(t) = e^{-1t}u(t) - e^{-2t}u(t)$$

$$h_a(t) = (e^{-t} - e^{-2t})u(t)$$

Step II : $t = nT_s$

$$h_a(nT_s) = (e^{-nT_s} - e^{-2nT_s})u(nT_s)$$

Step III

Apply Z-T

$$H(z) = \sum_{n=-\infty}^{\infty} h(nT_s)\, z^{-n}$$

$$= \sum_{n=0}^{\infty} \left(e^{-nT_s} - e^{-2nT_s}\right) \cdot z^{-n}$$

$$= \sum_{n=0}^{\infty} e^{-nT_s} z^{-n} - \sum_{n=0}^{\infty} e^{-2nT_s} z^{-n}$$

$$= \sum_{n=0}^{\infty} \left(e^{-T_s} z^{-1}\right)^{n} - \sum_{n=0}^{\infty} \left(e^{-2T_s} z^{-1}\right)^{n}$$

$$H(z) = \frac{1}{1 - e^{-T_s} \cdot z^{-1}} - \frac{1}{1 - e^{-2T_s} z^{-1}}$$

$$= \frac{z^{-1}\left(e^{-T_s} - e^{-2T_s}\right)}{\left(1 - e^{-T_s} z^{-1}\right)\left(1 - e^{-2T_s} z^{-1}\right)}$$

$$T_s = \frac{1}{f_s} = \frac{1}{5} = 0.2$$

$$H(z) = \frac{z^{-1}\left(e^{-0.2} - e^{-0.4}\right)}{\left(1 - e^{-0.2} z^{-1}\right)\left(1 - e^{-0.4} z^{-1}\right)}$$

After solving

$$\boxed{H(z) = \frac{0.148\, z}{z^2 - 1.48 z + 0.548}}$$

Exp2:- $H_a(S) = \dfrac{S+0.1}{(S+0.1)^2+9}$

Solution:-

Step I:- ILT

$$H_a(s) = \frac{S+0.1}{(S+0.1)^2+9}$$

Recall
$a^2-b^2 = (a+b)(a-b)$
$\rightarrow a^2-(jb)^2$
$a^2-j^2b^2$
a^2+b^2

$$= \frac{S+0.1}{(\underbrace{S+0.1}_{a} - \underbrace{j3}_{b})(\underbrace{S+0.1}_{a} + \underbrace{j3}_{b})}$$

Now PF

$$= \frac{A}{S+0.1-j3} + \frac{B}{S+0.1+j3}$$

$$A = 1/2 \quad , \quad B = 1/2$$

$\therefore$

$$H_a(S) = \frac{\frac{1}{2}}{S+(0.1-3j)} + \frac{1/2}{S+(0.1+3j)}$$

$$h_a(t) = \frac{1}{2}\left[e^{-(0.1-3j)t} + e^{-(0.1+3j)t}\right]u(t)$$

Step II:-
$$h(nT_s) = \frac{1}{2}\left[e^{-(0.1-3j)nT_s} + e^{-(0.1+3j)nT_s}\right]u(nT_s)$$

Step III Apply ZT

$$H(z) = \sum_{n=0}^{\infty} \frac{1}{2}\left[e^{-(0.1-3j)nT_s} + e^{-(0.1+3j)nT_s}\right] z^{-n}$$

$$H(z) = \frac{1}{2}\left[\sum_{n=0}^{\infty} e^{-(0.1-3j)nT_s} z^{-n} + \sum_{n=0}^{\infty} e^{-(0.1+3j)nT_s} z^{-n}\right]$$

$$= \frac{1}{2}\left[\frac{1}{1-e^{-T_s(0.1-3j)}z^{-1}} + \frac{1}{1-e^{-T_s(0.1+3j)}z^{-1}}\right]$$

$$H(z) = \frac{1-(e^{-0.1T_s}\cos(3T))z^{-1}}{1-(2e^{-0.1T_s}\cos 3T)z^{-1} + e^{-0.2T}z^{-1}}$$

-: Bilinear Transformation :-

$$H(s) = H(z)\Big|_{s = \left(\frac{1-z^{-1}}{1+z^{-1}}\right)\frac{2}{T}}$$

$$\Omega = \frac{2}{T}\tan\left(\frac{\omega}{2}\right)$$

Exp 1:- Convert the analog filter with system function

$$H(s) = \frac{s+0.1}{(s+0.1)^2+16}$$

into digital IIR filter using BLT. The digital filter should have resonant freq. $\omega_r = \pi/2$

Solution :-

T is missing

$$H(s) = \frac{(s+0.1)}{(s+0.1)^2+16}$$

Standard eq. is $\dfrac{s+a}{(s+a)^2+(\Omega_c)^2}$

$$\Omega_c = \frac{2}{T}\tan\left(\frac{\omega}{2}\right)$$

$$\Omega_c = \frac{2}{T}\tan\left(\frac{\pi}{4}\right)$$

$$\Omega_c = \frac{2}{4}\tan\left(\frac{\pi}{4}\right)$$

$$\boxed{\Omega_c = \frac{1}{2}\ \text{Sec}}$$

$$H(z) = H_a(s)\Big|_{s=\frac{2}{T}\left(\frac{1-z^{-1}}{1+z^{-1}}\right)}$$

$$= \frac{(s+0.1)}{(s+0.1)^2+16}\Bigg|_{s=\frac{2}{1/2}\left(\frac{1-z^{-1}}{1+z^{-1}}\right)}$$

$$H(z) = \frac{4\left(\frac{1-z^{-1}}{1+z^{-1}}\right)+0.1}{\left(4\left(\frac{1-z^{-1}}{1+z^{-1}}\right)+0.1\right)^2+16}$$

$$= \frac{\dfrac{4(1-z^{-1})+0.1(1+z^{-1})}{(1+z^{-1})}}{\dfrac{\left[4(1-z^{-1})+0.1(1+z^{-1})\right]^2+16(1+z^{-1})^2}{(1+z^{-1})^2}}$$

$$H(z) = \frac{\left[4(1-z^{-1})+0.1(1+z^{-1})\right](1+z^{-1})}{\left[4(1-z^{-1})+0.1(1+z^{-1})\right]^2+16(1+z^{-1})^2}$$

$$\boxed{H(z) = \frac{0.128z^2+0.006z-0.122}{z^2+0.0006z+0.975}}$$

Exp2 :- Design a single pole LPF with a 3 dB B.W of 0.2π using BLT approach applied to analog filter $H(s) = \frac{\Omega_c}{s+\Omega_c}$

here Ω_c is 3-dB B.W of analog filter.

Solution :- $\omega_c = 0.2\pi \Big|_{3dB}$

$$\Omega_c = \frac{2}{T}\tan\left(\frac{\omega}{2}\right)$$

$$= \frac{2}{T}\tan\left(\frac{0.2\pi}{2}\right)$$

$$\Omega_c = \frac{0.65}{T}$$

$$H(z) = H(s)\Big|_{s=\frac{2}{T}\left(\frac{1-z^{-1}}{1+z^{-1}}\right)}$$

$$H(z) = \frac{\Omega_c}{s+\Omega_c}\Big|_{s=\frac{2}{T}\left(\frac{1-z^{-1}}{1+z^{-1}}\right)}$$

$$H(z) = \frac{0.65/T}{\frac{2}{T}\left(\frac{1-z^{-1}}{1+z^{-1}}\right) + \frac{0.65}{T}}$$

$$H(z) = \frac{0.65}{2\left(\frac{1-z^{-1}}{1+z^{-1}}\right) + 0.65}$$

$$= \frac{0.65\left(1+z^{-1}\right)}{2(1-z^{-1}) + 0.65(1+z^{-1})}$$

$$H(z) = \frac{0.65 + 0.65z^{-1}}{2-2z^{-1}+0.65+0.65z^{-1}}$$

After solving

$$\boxed{H(z) = \frac{0.345(z+1)}{z-0.509}}$$